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CARTOGRASS 2026
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MAREA: a method for mapping intertidal topography using Sentinel-2 openEO data cubes

The tide arrives late: MAREA reads the delay from satellite images and corrects the maps for it.

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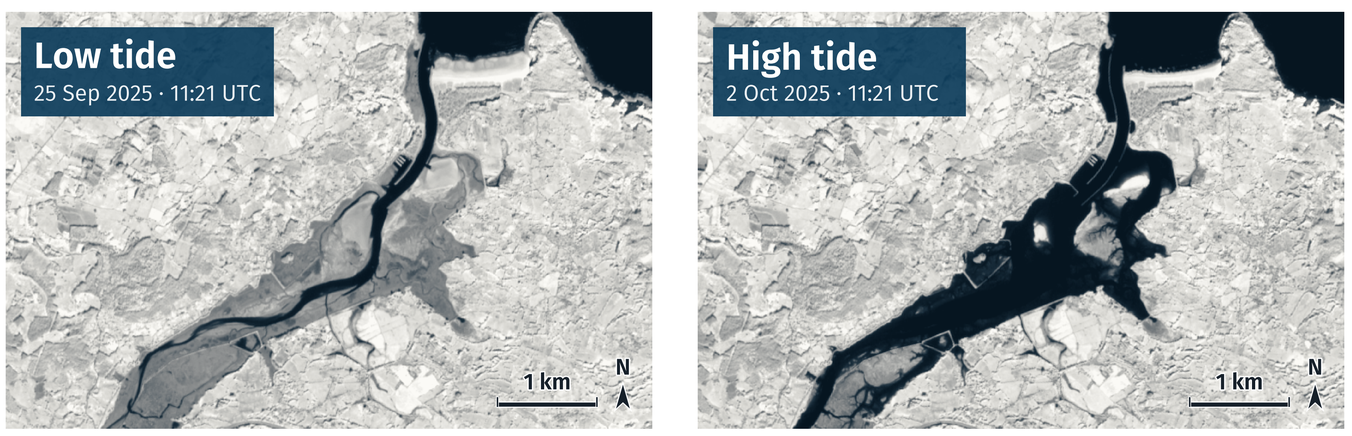
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Why emerged time?

A main driver of where *Zostera noltei* grows is **emerged time** (emersion time): the share of time the ground spends out of the water, set by its elevation and the local tide.

Photo: *Zostera noltei* exposed at low tide, Ría de Villaviciosa (authors' own)



Ría de Villaviciosa in Sentinel-2 near-infrared: water is dark. Contains modified Copernicus Sentinel data (2025).

Two gaps

Unknown elevation. Most tidal flats lack an up-to-date elevation map.

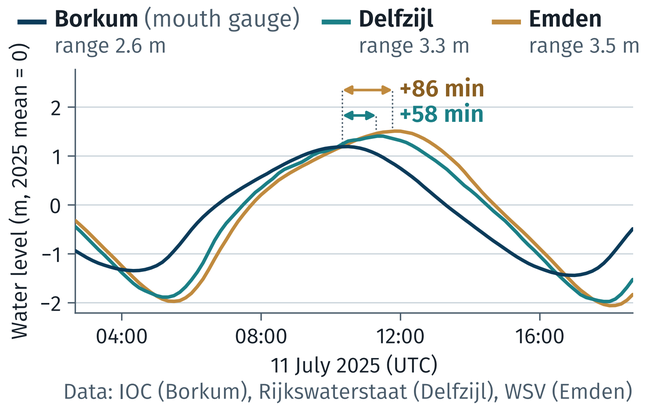
A late tide. Global tide models give the tide at the open coast; inside estuaries and tidal basins it arrives up to **1.5–2 hours** later, so maps made with them misjudge when flats flood.

MAREA addresses both with the Sentinel-2 archive.

What does a satellite see of the tide?

Mostly wet or dry: a pixel is wet when the water stands above its ground [1, 2]. Seen many times, at different tide stages, each pixel becomes a **yes-or-no tide gauge**.

If the tide arrives late, the pixel floods late [2].



Data: IOC (Borkum), Rijkswaterstaat (Delfzijl), WSV (Emden)
Ems estuary: inside, high water arrives later and higher.

OBJECTIVE

Can the Sentinel-2 archive measure, section by section along the channel, how late the tide arrives and, with that delay corrected, map intertidal elevation and emerged time?

Why decimeters matter: around mean sea level, a flat 30 cm higher spends about 1 hour more out of the water per tide (Gijón tide gauge, 2024).

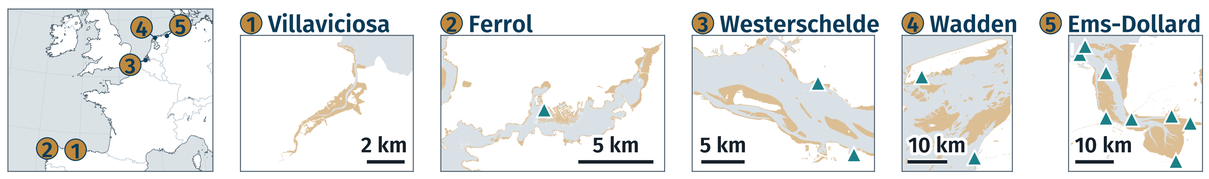
Pilot result

Ría de Villaviciosa, 194 precise GPS points: typical elevation error **0.19 m** (offset removed), 6% of the mean spring range. Little delay there: a test of the elevation step.

QR
Validation details

Materials and methods

Study sites and data



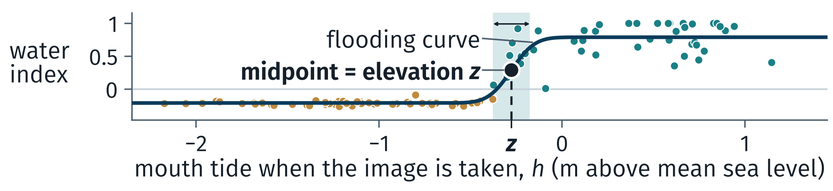
intertidal flats water ▲ tide gauge used as a check. Two rias in Spain; two North Sea estuaries and a Wadden Sea tidal basin in the Netherlands and Germany. Basemap: Natural Earth.

Input data (all MAREA needs: no field surveys): INPUT 1 Sentinel-2 images, 10 m, 2023–2025, as openEO data cubes in the Copernicus cloud [3]. INPUT 2 Tide at the mouth: global tide model EOT20 [4] and, at four sites, tide gauges at the mouth.

Independent checks (never used to make the maps): ① precise GPS survey (RTK-GNSS) and PNOA airborne LiDAR ② a tide gauge in the inner ria ③④⑤ Vaklodingen depth surveys and tide gauges inside each site

A pixel is a tide gauge

the method's step ④ on one real pixel, Ría de Villaviciosa, 2023–2025



- **wet**: index above 0
- **dry**: index 0 or below
- width = uncertainty of *z*

wet when *h* > *z*
3 km from the mouth: delay τ ≈ 0

How MAREA works: follow one pixel

Schematic: one imaginary pixel

h(*t*) tide at the mouth at time *t* (m above mean sea level) · τ delay behind the mouth (min; fitted per 3 km of channel) · *z* ground elevation of the pixel (m, same scale as *h*)

INPUT 1 · IMAGES

① Seen many times
Sentinel-2 data cube
time · wet · cloud · dry
10-m pixels · 2023–2025
each look: wet or dry
1 look = 1 cloud-free view
water index from green and near-infrared light

+

INPUT 2 · TIDE AT THE MOUTH

② Its tide arrives late
mouth *h*(*t*) · Ems-Dollard · pixel · its 3-km section *h*(*t* − τ)
no delay: wet · seen dry · ground *z* · τ
image at time *t*
wet when *h*(*t* − τ) > *z*
the mouth tide reaches its section τ later
MAREA shifts the tide in time, not height

③ Its section's delay
delays τ tried (minutes, examples)
• **dry** • **wet**: one look of this pixel
ground *z* · fit
0 mixed
45 mixed
90 clean
135 mixed
its water level, *h*(*t* − τ) →
best delay: cleanest split
fit: judged on its section's pixels

④ Its elevation
water index · width · flooding curve · midpoint · dry · wet · *z*
its water level, *h*(*t* − τ)
midpoint = elevation *z*
width = uncertainty of *z*
needs wet and dry looks, otherwise left empty

⑤ Its emerged time
its water level, *h*(*t* − τ) · share of its time
z · 4 days →
61% under water · **39%** out of the water
emerged time *E*(*z*) = share of time its water is below *z*: *h*(*t* − τ) < *z*
at all times, not only image times

OUTPUT

⑥ All pixels: the maps
elevation low – high
uncertainty low – high
emerged time 0 – 100%
base layer for the potential habitat of *Zostera noltei*
ecologists set the habitat limits

Results

Conclusions

Pending validation

[pending] Delay from images vs tide gauges, four sites

[pending] What the delay correction changes in elevation, site by site

[pending] What the emerged-time map shows for *Zostera noltei*

Acknowledgments
[To be supplied by the authors.]

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Data
Contains modified Copernicus Sentinel data (2023–2025; Villaviciosa site map 2016–2025). Tide gauges: IOC Sea Level Station Monitoring Facility, Rijkswaterstaat, WSV, Puertos del Estado. Bathymetry: Rijkswaterstaat (Vaklodingen). LiDAR: PNOA-IGN. Tide model: EOT20.

References
[1] Bishop-Taylor et al. (2019) *Estuar. Coast. Shelf Sci.* 223, 115–128.
[2] Granadeiro et al. (2021) *Remote Sens.* 13(2), 320.
[3] Schramm et al. (2021) *Remote Sens.* 13(6), 1125.
[4] Hart-Davis et al. (2021) *Earth Syst. Sci. Data* 13, 3869–3884.

Code
github.com/coastal-AI/Intertidal_analysis